

Generation Interconnection Request System Impact Study Report

For Queue Project Q20-1542

Mccabe 138 kV
May 16, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1542. The Interconnection Customer proposes a 267.6 MW solar photovoltaic facility interconnecting at the McCabe 138 kV substation in the North Carolina area. The request was received on August 04, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 424 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$8,154,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: SORENSEN - MEYER RD 138 kV L
Contingency: N TOWER TAP - MCCARTNEY 345 kV L
Loading: 107.6% (Rating: 221 MVA)
DFAX: 5.1%
Pre-project loading: 101.4%
2. Facility: BEAR - FLUVANNA P.S 345 kV Line
Contingency: LEAVITT SUBSTATION - BIRCHWOOD 345 kV
Loading: 103.5% (Rating: 1,663 MVA)
DFAX: 15.9%
Pre-project loading: 100.9%
3. Facility: DIST 1000 - BIRCHWOOD 138 kV
Contingency: BEAR ISLAND - DUNN SIDING 345 kV
Loading: 103.1% (Rating: 515 MVA)
DFAX: 5.6%
Pre-project loading: 100.2%
4. Facility: TRENTON DIESEL - DIST 1000 138 kV
Contingency: BEAR ISLAND - DUNN SIDING 345 kV Line
Loading: 103.1% (Rating: 527 MVA)
DFAX: 5.6%
Pre-project loading: 100.2%
5. Facility: REDBED PLAINS WIND FARM - WESTCOAST CELLUFIBRE 138 kV L
Contingency: MITCHELL DAM - SKAGGS ISLAND JCT 2 345 kV
Loading: 100.5% (Rating: 457 MVA)
DFAX: 6.0%
Pre-project loading: 97.0%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
SORENSEN - MEYER RD 138 kV	Reconductor line	\$1,085,000	23% (\$249,000)
BEAR - FLUVANNA P.S 345 kV Line	Reconductor line	\$46,572,000	10% (\$4,558,000)
DIST 1000 - BIRCHWOOD 138 kV L	Reconductor line	\$10,469,000	12% (\$1,290,000)
TRENTON DIESEL - DIST 1000 138 kV	Reconductor line	\$15,555,000	12% (\$1,916,000)
REDBED PLANS WIND FARM - WESTCOAST CELLUFIBRE 138 kV	Reconductor line	\$1,124,000	13% (\$141,000)
	Total allocated to project		\$8,154,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.